

Komalpreet Kaur

AI/ML Engineer

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Objective

AI/ML Engineer building LLM inference systems, agentic AI, and multimodal AI across speech, vision, and memory. Focused on shipping real, working systems — still learning, always building. Open to roles across startups and established companies.

Projects

Chimera Python, PyTorch, FastAPI, Transformers, CUDA

LLM inference engine built from scratch

- Built a GPT-2 inference engine from scratch (attention, KV caching, sampling); validated to 5e-5 logit accuracy vs HuggingFace.
- Implemented KV caching for 46x faster decode steps and continuous batching running 8 concurrent requests per forward pass.
- Increased throughput 3.7x from 9.8 to 36.2 tokens/sec and cut time-to-first-token from 25.8s to 3.4s, with per-token telemetry.
- Repository Link: <https://github.com/Komalpreet2809/Chimera>

Soma (soma.komalpreet.me)

FastAPI, React, LangChain, Neo4j, ChromaDB, SQLite, Docker

Cognitive memory architecture

- Designed a 4-layer memory system that recalls per-user facts across sessions in under 100 ms per query.
- Built a 3-stage cognitive pipeline fusing knowledge-graph and vector recall, streamed live with reasoning telemetry.
- Converted raw chat into a clean knowledge graph via LLM extraction, with an automated sleep cycle that prunes memory.
- Repository Link: <https://github.com/Komalpreet2809/SOMA>

Vanta (vanta.komalpreet.me)

PyTorch, SpeechBrain, FastAPI, Next.js, Docker, Hugging Face

Speaker-conditioned voice extraction

- Isolates a target voice from overlapping speech using a short reference clip, which blind denoisers cannot separate.
- Built and trained both networks from scratch on one 8 GB laptop GPU: a Conv-TasNet separator and ECAPA-TDNN speaker encoder.
- Achieved +9.28 dB median SI-SDR on unseen speakers, outperforming the pretrained encoder it replaced on 3 of 4 benchmarks.
- Repository Link: <https://github.com/Komalpreet2809/Vanta>

Specula (specula.komalpreet.me)

Python, FastAPI, PyTorch, HuggingFace, OpenCV

Explainable deepfake detection toolkit

- Detects AI-generated images and shows why — heatmaps point to exact regions, a problem black-box classifiers can't solve.
- Fuses 5 forensic signals into a 0–100 verdict, with confidence-adaptive weighting scaling the classifier from 40% to 75%.
- Live in production — upload an image, get a full forensic breakdown with 10-image batching and PDF reports.
- Repository Link: <https://github.com/Komalpreet2809/Specula>

Skills

LLM & Generative AI:	LangChain, LangGraph, RAG, Vector DBs, Knowledge Graphs, Prompt Engineering
AI Systems & Infrastructure:	Tokenization, KV Caching, Continuous Batching, Model Serving, GPU Inference, CUDA
Machine Learning:	PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, SpeechBrain, NLP
Programming:	Python, Java, C++, Bash, SQL
Backend & APIs:	FastAPI, REST APIs, Model Deployment, WebSockets
Frontend:	React, Next.js, JavaScript, HTML, CSS
Databases:	ChromaDB, Neo4j, SQLite, NoSQL
Automation & Tools:	Playwright, Docker, Git, GitHub Actions, CI/CD, Linux
Cloud & Deployment:	AWS, Azure, Hugging Face Spaces, Vercel

Education

Lovely Professional University

2023 – 2027

Bachelor of Technology - Computer Science and Engineering — CGPA: 8.56
Punjab

Phagwara,

Shri Rajendra Giri Memorial Academy

2021 – 2022

12th — Percentage: 93%

Kheri, Uttar Pradesh

Shri Guru Nanak Dev Public School

2019 – 2020

10th — Percentage: 96.2%

Kheri, Uttar Pradesh

Note: Links to projects are embedded in their titles. View full portfolio at komalpreet.me